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Low-Complexity Hybrid Precoding in Massive Multiuser MIMO Systems

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Abstract

Massive multiple-input multiple-output (MIMO) is envisioned to offer considerable capacity improvement, but at the cost of high complexity of the hardware. In this paper, we propose a low-complexity hybrid precoding scheme to approach the performance of the traditional baseband zero-forcing (ZF) precoding (referred to as full-complexity ZF), which is considered a virtually optimal linear precoding scheme in massive MIMO systems. The proposed hybrid precoding scheme, named phased-ZF (PZF), essentially applies phase-only control at the RF domain and then performs a low-dimensional baseband ZF precoding based on the effective channel seen from baseband. Heavily quantized RF phase control up to 2 bits of precision is also considered and shown to incur very limited degradation. The proposed scheme is simulated in both ideal Rayleigh fading channels and sparsely scattered millimeter wave (mmWave) channels, both achieving highly desirable performance.

Keywords

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